## Binary Visual–Verbal Learning Style Scoring

For each learner, a binary Visual–Verbal preference is determined from the learner's aggregated interaction counts. The interactions are grouped into Visual and Verbal categories based on the predefined feature mapping.

### 1. Visual and Verbal Interaction Scores

For learner i, the Visual interaction score is calculated as:

$V_i = \text{Video}_i + \text{Access}_i$

The Verbal interaction score is calculated as:

$R_i = \text{Wiki}_i + \text{Discussion}_i$

### 2. Binary Preference Score

The relative preference of a learner is represented using the following normalized score:

$S_i = (V_i - R_i) / (V_i + R_i)$

The score ranges from −1 to +1. Positive values indicate greater Visual interaction, whereas negative values indicate greater Verbal interaction.

### 3. Binary Label Assignment

For binary classification, the learner is assigned to the style with the higher interaction count:

$\text{Label}_i = \text{Visual}, \text{ if } V_i > R_i$

$\text{Label}_i = \text{Verbal}, \text{ if } V_i < R_i$

When $V_i = R_i$, a predefined tie-breaking rule should be applied to retain the two-class setting. For example, ties can be assigned to Verbal.

### 4. Interpretation

Thus, the binary label represents the learner's dominant interaction tendency based on the aggregated behavioral counts. This rule-based label can subsequently be compared with the clusters obtained from unsupervised clustering to examine whether the behavioral clusters correspond to Visual- or Verbal-dominant learners.